WORKSHEET

Modelling arithmetic and geometric sequences

1 In the construction of a major road, 57 m of road is completed in the first month, and 78 m is completed every month after that.

a How much road will have been constructed after 1 year?

b If the road takes 3 years and 8 months to build, what will be the length of the road once completed?

2 A section of fence is built on a slope. The lengths of the pieces of timber in this part of the fence form an arithmetic sequence. The shortest piece is 0.8 m and the longest is 2 m. There are 61 fence posts in total.

a Calculate the difference in length between adjacent fence posts.

b Find the total length of timber required to make the fence.

3 Marcia's salary increases by a set amount annually. In 2011, her salary was $102 900, and in 2014 her salary is $110 700.

a By how much does her salary increase each year?

b From the start of 2010 until the end of 2020, what will be her total earnings from this job?

4 A painting appreciates (increases in value) by 16% p.a. It is currently worth $20 000.

a How much will it be worth in 2 years?

b How much will it be worth in 11 years?

c In how many years will it be worth more than $50 000?

© Cengage Learning Australia Pty Ltd 2014

5 As a result of deforestation, the koala population in Port Meribula is decreasing at a rate of 12.5% per year. In 2013, the population was 2455.

a What will be the koala population in 2015 in Port Meribula? Answer to the nearest whole number.

b In what year will the koala population fall below 1000?

6 In the opening of a new theme park it was decided to limit the number of entries over the first 2 weeks while staff became accustomed to their new jobs. Each day, an extra 50 customers are admitted access to the park, with a total of 725 being admitted on the 14th day of operation.

a How many people were admitted to the park on the first day?

b How many people were admitted in total over the first 2 weeks of the theme park opening?

7 A waterslide is mounted on posts that are two metres apart. The first post is 30 m tall. Each subsequent post is 70% of the height of the previous post.

a What is the height of the tenth post, to the nearest centimetre?

b Between which two posts is the height of the slide half the original height?

8 A tree grows to a height of 3 m in its first year. In each subsequent year, the tree grows $\frac{3}{4}$ of the previous year's growth.

a To the nearest centimetre, how much will the tree grow in its fourth year?

b To the nearest centimetre, how tall will the tree be when it is 10 years old?

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** 915 m

b 3411 m

2 **a** 0.02 m

b 85.4 m

3 **a** $2600

b $1 246 300

4 **a** $26 912

b $102 345.29

c 7 years

5 **a** 1880

b 2020

6 **a** 75

b 5600

7 **a** 8 cm

b The second and the third

8 **a** 1.27 m

b 11.32 m

© Cengage Learning Australia Pty Ltd 2014